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Title:

MQTT Data Subscription and Debug Display using Node-RED

Objective:

To subscribe DHT22 temperature data published by ESP32 using MQTT and display it in Node-RED Debug window.

Tools / Software Used:

- ESP32 (Simulator in Wokwi / VS Code)
- DHT22 sensor
- Mosquitto MQTT Broker (running on port 1883)
- Node-RED (<http://localhost:1880>)

Procedure (Step-by-Step):

1. Start Mosquitto Broker:

Open terminal / command prompt and run:

2. mosquitto -v

Output should confirm:

mosquitto running

3. Run ESP32 Code (Publisher):

- Make sure ESP32 code has correct broker IP:
- `const char* mqtt_server = "";`
- ESP32 publishes temperature to topic:
- `home/lab1/temp`
- Serial monitor shows:
- MQTT connected
- Publishing temperature: 26.5

4. Start Node-RED:

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5. node-red

Open browser: <http://localhost:1880>

6. **Add MQTT-in Node:**

- Drag **MQTT-in node**
- Configure:
 - Server: localhost
 - Port: 1883
 - Topic: home/lab1/temp
 - QoS: 0
 - Action: Subscribe to single topic
- Click **Done**

The screenshot shows the 'Edit mqtt in node' configuration window in Node-RED. The window has a title bar 'Edit mqtt in node' and three buttons at the top: 'Delete', 'Cancel', and 'Done'. Below the buttons is a 'Properties' tab with a settings icon, a document icon, and a preview icon. The configuration fields are as follows:

- Server:** A dropdown menu showing 'test.mosquitto.org:1883' with a pencil icon and a plus icon to its right.
- Action:** A dropdown menu showing 'Subscribe to single topic' with a downward arrow.
- Topic:** A text input field containing 'home/lab1/temp'.
- QoS:** A dropdown menu showing '0' with a downward arrow.
- Output:** A dropdown menu showing 'auto-detect (parsed JSON object, string or buff)' with a downward arrow.
- Name:** A text input field containing 'Name'.

At the bottom left, there is a checkbox labeled 'Enabled' which is currently checked.

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7. Add Debug Node:

- Drag **Debug node**
- Output: msg.payload
- Connect: MQTT-in → Debug

The screenshot shows the 'Edit debug node' dialog box. At the top, there are three buttons: 'Delete', 'Cancel', and 'Done'. Below these is a 'Properties' section with a gear icon and three sub-panels: 'Output', 'To', and 'Name'. The 'Output' panel has a dropdown menu showing 'msg. payload'. The 'To' panel has three checkboxes: 'debug window' (checked), 'system console' (unchecked), and 'node status (32 characters)' (unchecked). The 'Name' panel has a text input field containing 'debug 1'. At the bottom of the dialog, there is a checkbox labeled 'Enabled' which is currently unchecked.

8. Deploy Flow:

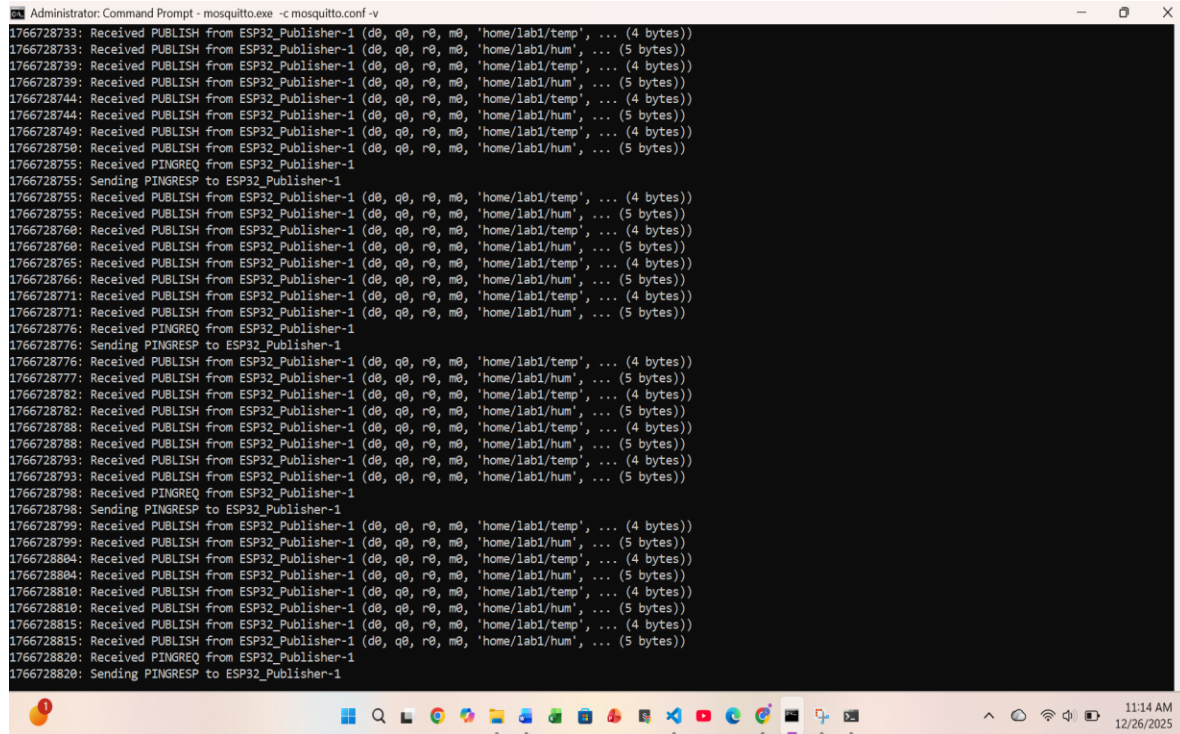
- Click **Deploy**
- Open **Debug tab** (right panel)
- You should see temperature values like:
- 26.4

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- 27.0

9. Screenshots:

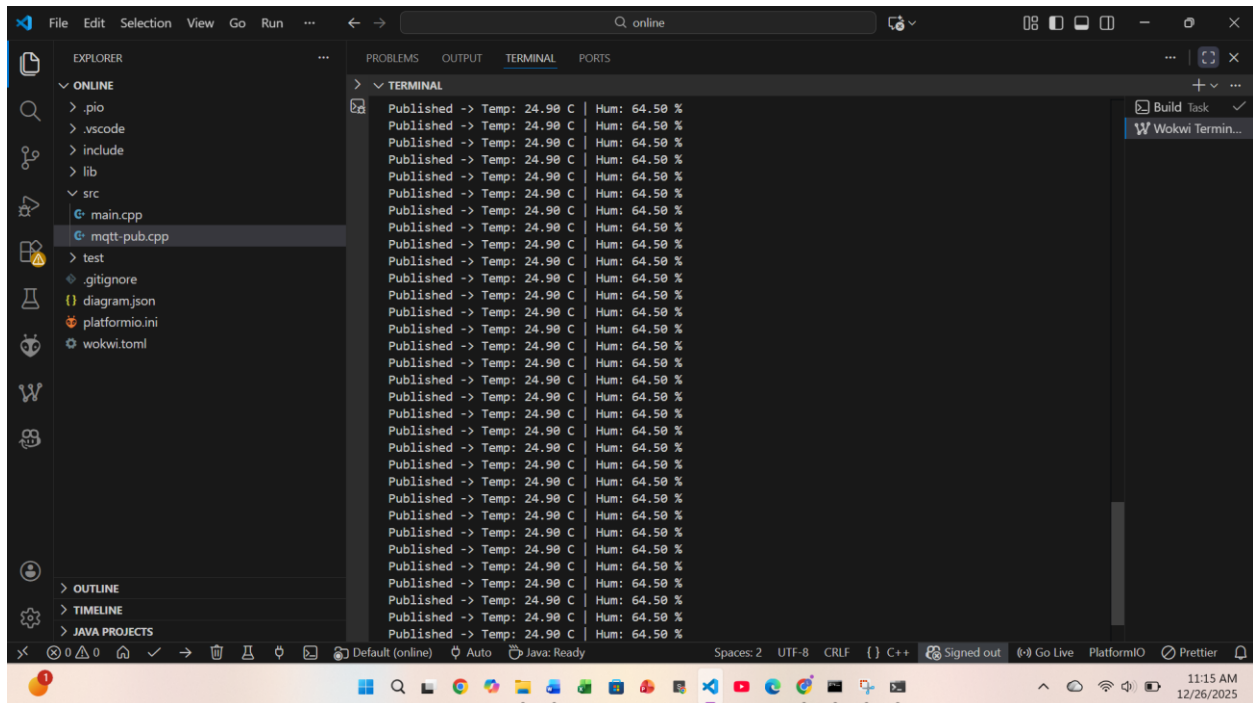
- Mosquitto running terminal



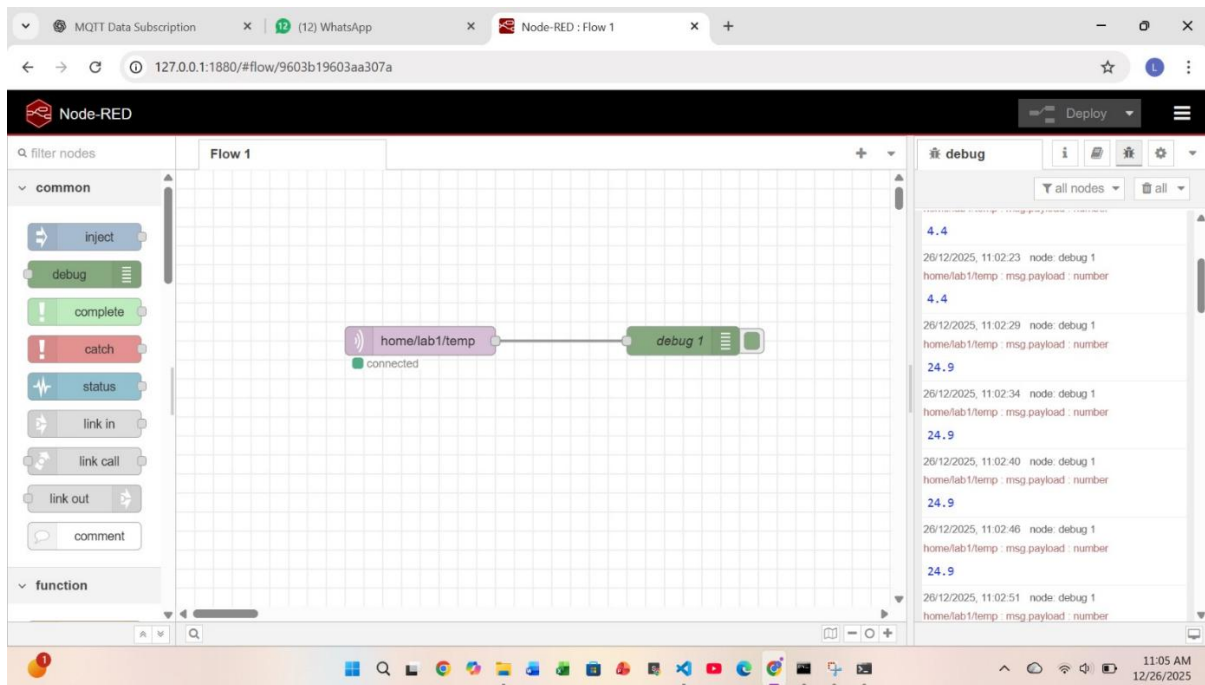
```
Administrator: Command Prompt - mosquitto.exe -c mosquitto.conf -v
1766728733: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728733: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728739: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728739: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728744: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728744: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728749: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728750: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728755: Received PINGREQ from ESP32_Publisher-1
1766728755: Sending PINGRESP to ESP32_Publisher-1
1766728755: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728755: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728760: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728760: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728765: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728766: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728771: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728771: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728776: Received PINGREQ from ESP32_Publisher-1
1766728776: Sending PINGRESP to ESP32_Publisher-1
1766728776: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728777: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728782: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728782: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728788: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728788: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728793: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728793: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728798: Received PINGREQ from ESP32_Publisher-1
1766728798: Sending PINGRESP to ESP32_Publisher-1
1766728799: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728799: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728804: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728804: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728810: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728810: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728815: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/temp', ... (4 bytes))
1766728815: Received PUBLISH from ESP32_Publisher-1 (d0, q0, r0, m0, 'home/lab1/hum', ... (5 bytes))
1766728820: Received PINGREQ from ESP32_Publisher-1
1766728820: Sending PINGRESP to ESP32_Publisher-1
```

- ESP32 serial output (Wokwi)

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- Node-RED flow (MQTT-in → Debug)



Conclusion:

ESP32 successfully published DHT22 temperature data using MQTT protocol. Node-RED subscribed to the topic home/lab1/temp through a local Mosquitto broker. The received data was displayed in the Debug window, demonstrating successful IoT data monitoring.